Note:

Click here to download the full example code

Plotting contour subplots with different projections

In this recipe, we will plot the same data using different projections as subplots to illustrate visually some available possibilities.

1. Import cf-python and cf-plot:

```
import cfplot as cfp

import cf
```

2. Read the field in:

```
f = cf.read("~/recipes/ggap.nc")[0]
```

3. List the projection types to use. Here we are using Cylindrical/Default, North Pole Stereographic, South Pole Stereographic, Mollweide, Mercator and Robinson. However there are several other choices possible, see: https://ncas-cms.github.io/cf-plot/build/user_guide.html#appendixc. Our chosen list is:

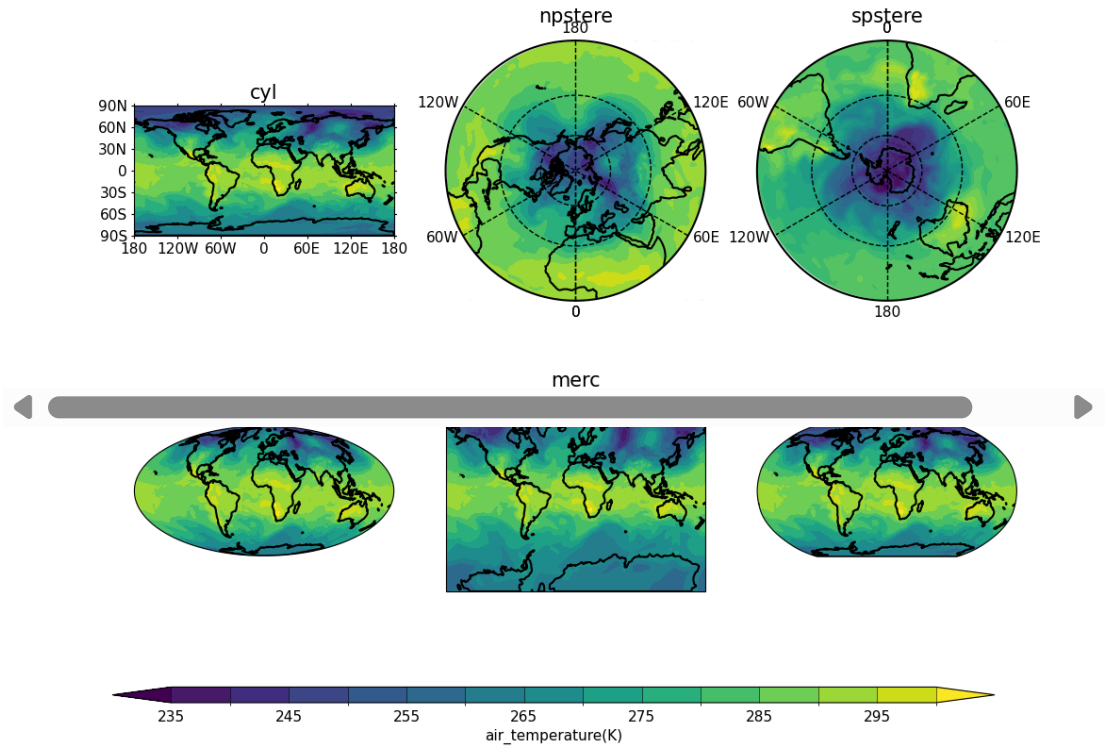
```
projtypes = ["cyl", "npstere", "spstere", "moll", "merc", "robin"]
```

4. Create the file with subplots. If changing the number of subplots, ensure the number of rows * number of columns = the number of projections. Here we are doing 6 projections so 2 x 3 is fine. Then loop through the list of projection types and plot each as a sub-plot:

```
cfp.gopen(rows=2, columns=3, bottom=0.2)
for i, proj in enumerate(projtypes):
    # gpos has 1 added to the index because it takes 1 as its first val
    cfp.gpos(i + 1)
    cfp.mapset(proj=proj)

    # For the final plot only, add a colour bar to cover all the sub-pl
    if i == len(projtypes) - 1:
        cfp.con(
            f.subspace(pressure=850),
            lines=False,
            title=proj,
            colorbar_position=[0.1, 0.1, 0.8, 0.02],
```

```
            colorbar_orientation="horizontal",
        )
    else:
        cfp.con(
            f.subspace(pressure=850),
            lines=False,
            title=proj,
            colorbar=False,
        )
```



Total running time of the script: (0 minutes 5.194 seconds)

Download Python source code: plot_16_recipe.py

Download Jupyter notebook: plot_16_recipe.ipynb

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